

PEDRO BARBOSA

Machine Learning / Platform Engineer — Python, SQL, Pipelines

Rio de Janeiro, Brazil · barbosamaverickv8@gmail.com · github.com/Amadeus-22 · amadeus-22.github.io/portfolio
English (fluent) · Portuguese (native) · German (basic)

SUMMARY

Engineer with a data and machine learning background (Computer Engineering, ML track, GPA 8.7) who builds and operates production systems. Day to day I work in Python and SQL on pipelines, model-driven research workflows and the infrastructure around them: reproducible runs, versioned schemas, containerised execution, CI, and the tests and logs that make failures visible instead of silent. I care about the boring part of ML platforms — reproducibility, correctness and observability — because that is where models quietly stop being right.

EXPERIENCE

Quantitative Developer — QuantConnect (freelance, remote)

September 2025 – present

- Python workflows on the LEAN engine: feature and factor research over large time-series datasets, model training and evaluation, batch backtesting over years of history.
- Machine learning models integrated into factor portfolios; systematic risk neutralisation (PCA) and portfolio construction through convex optimisation.
- Reproducibility as a requirement: deterministic runs, exact numerical handling, versioned experiments, and detection of data anomalies (gaps, misaligned timestamps, survivorship bias) that otherwise corrupt results.

Full Stack Developer — TIIX (remote)

April 2026 – present

- One of two developers on a healthcare platform in production, used daily by patients, clinics and professionals.
- Build and maintain backend services end to end: relational modelling and versioned migrations (23 of the repository's 50), REST endpoints, and the largest share of the automated test suite.
- Operate what I ship: investigate logs, database state and API responses, find root causes, resolve incidents, and document cause and solution — over a thousand written change records.
- Deliver through Git-based workflows with code review and CI pipelines.

Python Developer — Freelance via Workana (remote)

January 2025 – March 2026

- Built data pipelines in Python (pandas, openpyxl) replacing manual spreadsheet processes across fleet, HR and inventory data, removing a recurring source of error.
- Delivered dashboards and reporting on top of those pipelines for operational decision making.
- Django systems with role-based access control and normalised relational schemas designed to stay fast as data grows.

Software Development Intern — TT&T Soluções em Informática (remote)

August 2024 – January 2025

- Django backend and MySQL modelling; unit tests with pytest; optimisation of slow queries and requests; integration with external APIs.

SELECTED WORK

QuantStrata — risk and exposure engine in Python with exact rational arithmetic, built so incomplete input can never be reported as zero risk: it refuses to produce a metric it cannot support. 70 test cases running in a containerised pipeline (Docker, GitHub Actions), merged through formal code review.

iod — open source daemon in Go: executes external jobs under a versioned contract with bounded timeouts, response validation, retry with backoff and orphan-job recovery; SQLite storage with numbered migrations. 5,100 lines of code covered by 5,675 lines of tests.

SKILLS

Languages: Python, SQL, Go, JavaScript, TypeScript, PHP, Bash

ML and data: scikit-learn, pandas, NumPy, feature engineering, model evaluation, experiment reproducibility, time-series datasets

Databases: PostgreSQL, MySQL, SQLite — modelling, queries, versioned migrations

Platform: Docker, AWS EC2, Linux, systemd, Git, GitHub Actions, GitLab CI

Engineering: clean code, automated testing (unit and integration), code review, maintainable system design, logging and troubleshooting in production

Learning next: PySpark, Databricks, Kubernetes and Terraform — the tools in this stack I have not used professionally yet

EDUCATION AND CERTIFICATIONS

Computer Engineering (Data and Machine Learning) — Centro Universitário IBMR · 2023 – 2028 (expected) · GPA 8.7/10

Machine Learning certificate and scientific initiation in Economic Mathematics — IMPA, 2024

Coursera: Python for Data Science and AI · HackerRank: Problem Solving (Intermediate), REST API (Intermediate), Python